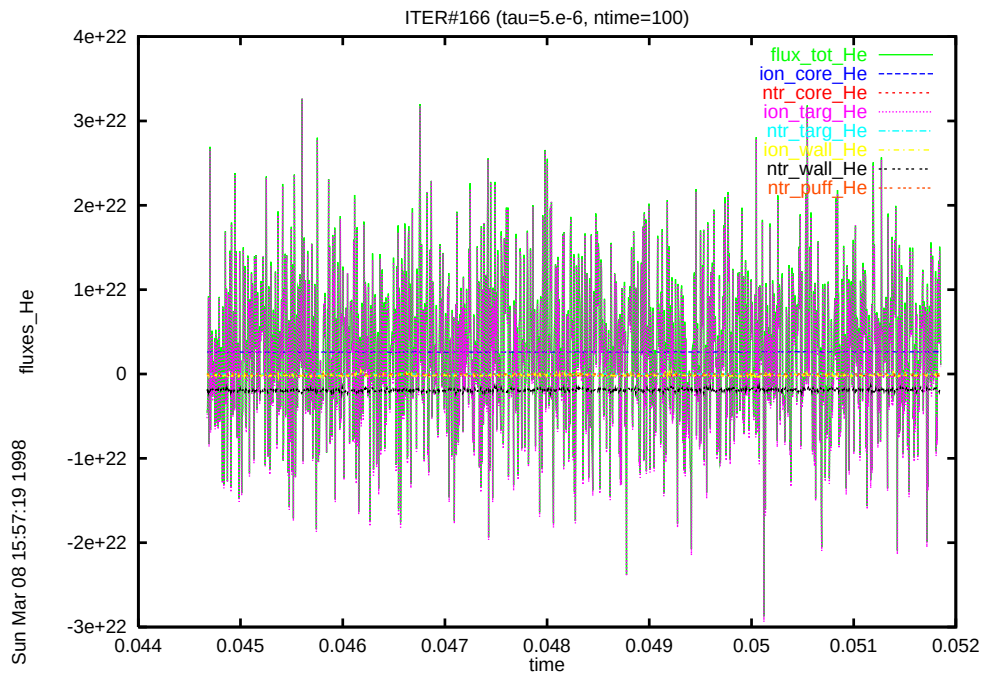

PARTICLE BALANCE IN B2-EIRENE

- For each particle species, each “stratum”, distinguish
 - Total ion source in B2, S^* :
 - volume integral of *positive* plasma sources for recombination
 - volume integral of *all* plasma sources for other sources
 - Total neutral sources in Eirene, I^* :
 - plasma fluxes onto the surfaces for recycling
 - constant influx for gas puff
 - volume integral of *negative* plasma sources for recombination
 - They are in balance after a call to Eirene (before a B2 step), and after B2 step they evolve differently
 - disbalance of order of percents of S^* , I^*
 - *but this can be an order of magnitude stronger than flux to the pump*
 - Introduce 2 scaling factors, α and λ , so that
 - give B2 (before iteration)
$$\tilde{S} = \alpha S^*, \quad \tilde{S}_{mom} = \lambda S_{mom}^*, \quad \tilde{S}_E = \lambda S_E^* (\Gamma = \lambda \Gamma^*)$$
 - get from B2 (after i -th iteration)
$$I(\alpha, i), \quad S(\alpha, i)$$
 - Let $\lambda_i = I(\alpha, i) / I^*$ – i. e., neutral fluxes and neutral-related sources of momentum and energy $\propto$ to the neutral particle source
 - Find α from the equation
$$\frac{S(\alpha, i)}{I(\alpha, i)} = \frac{S^*}{I^*}$$
solving this equation in the course of the general iterations in each step of B2
 - *Such a procedure allows to reduce the particle disbalance by ~ 3 orders of magnitude, making it possible to study the effects related to rather minor flows from the gas puff and to the pump*
-

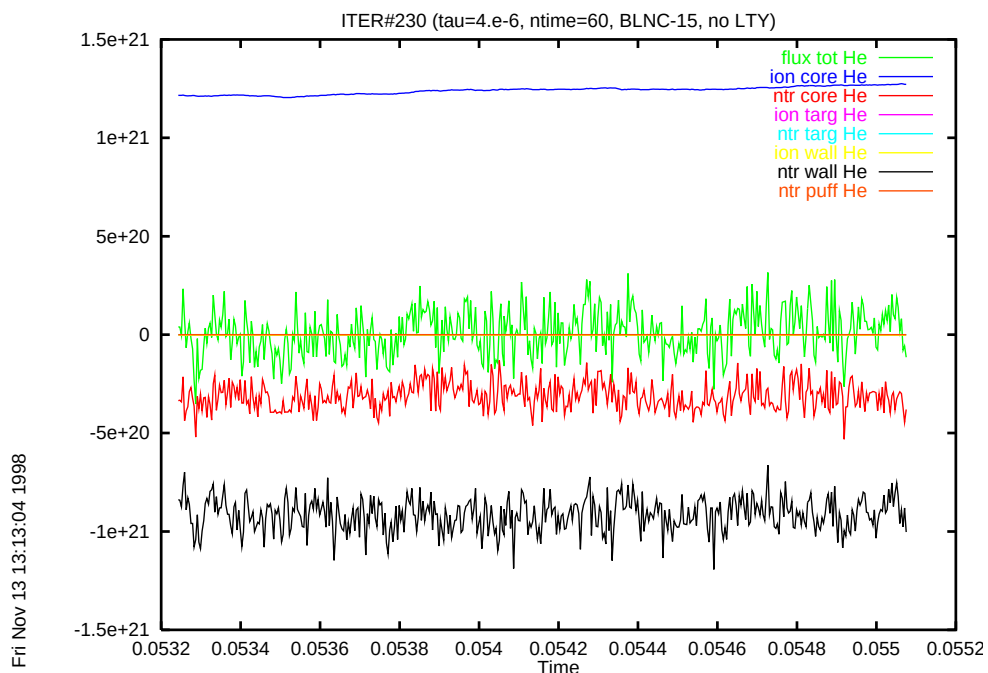
- Two examples of the effect of this correction are shown below.
 - Time traces of different terms in the particle balance are plotted
 - values taken after B2 steps
 - **green** line shows the total disbalance
 - **magenta** is the noise from plasma at the targets
 - **black** is the flow to the pump
 - Two different cases are shown

• Helium balance

- without source correction:

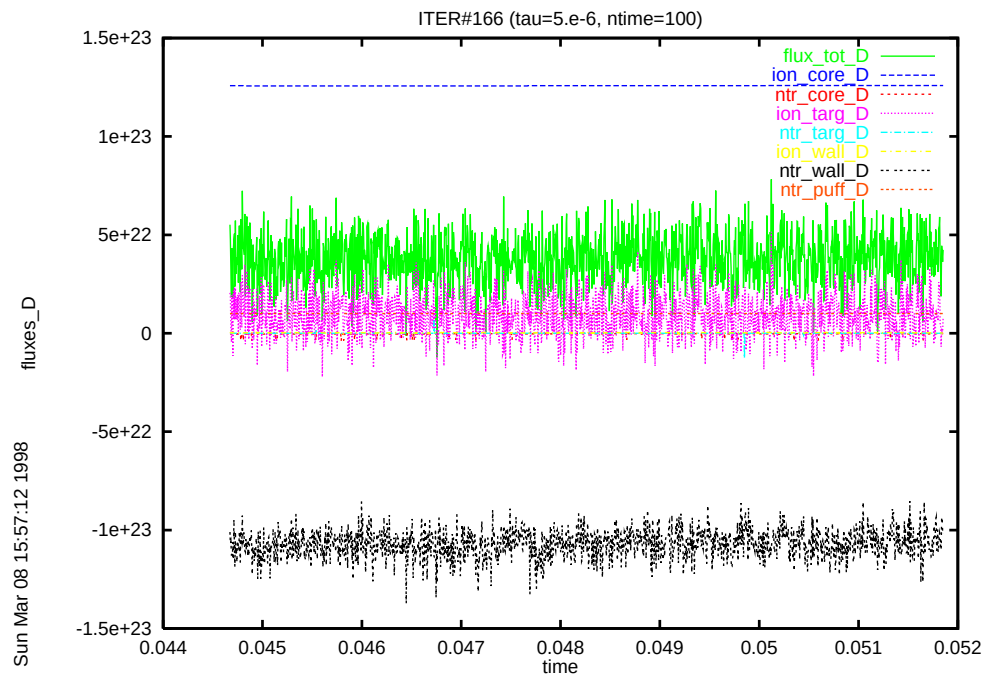


- with source correction:

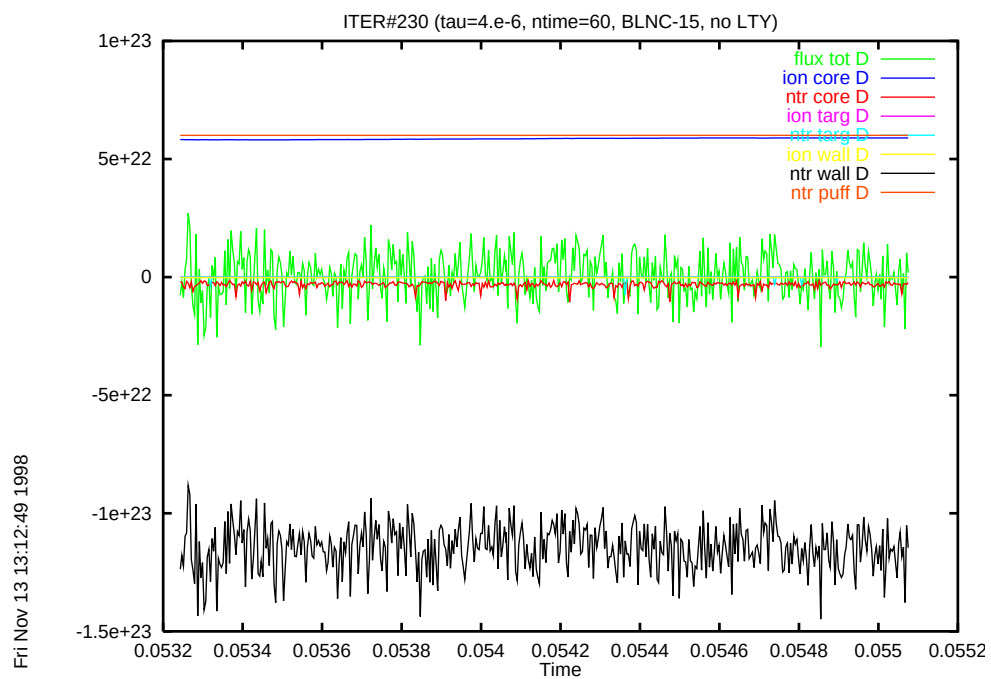


- Deuterium balance

- without source correction:



- with source correction:



The following should be set up in the input files in order to have the particle balance corrections

- **B2** (b2.parameters, section "&NEUTRALS"):

Name	Legend	Default
lstrasc1	Species "j" in stratum "i" is to be scaled with the species $lstrasc1(i, j)$. For recycling species (D, He, Ne, ...) should be equal to the species sequence number; for the sputtered species (C) it is more reasonable to specify the hydrogen species here. 0 means no correction at all, <0 means scaling simply proportional to the impinging ion flow (the old version).	-1, ..., -1
lmolscl	Molecule "j" is to be scaled with the species $lmolscl(j)$.	1, 0, ..., 0
lsns	List of "non-default standard surfaces" from Eirene to be used for calculation of the neutral fluxes. Should be in the same order as the strata are listed in "b2.parameters" file, but multiple surfaces can be specified for the same stratum.	0, ..., 0
ksns	Numbers of items in "lsns" for each stratum	0, ..., 0
mlcmp	Molecule of kind "j" contains $mlcmp(i, j)$ atoms of kind "i"	2, 0, ..., 0
gpfc	$gpfc(i, j)$ is a relative fraction of the nuclei of kind "i" in a gas-puffing source of stratum "j"	1, 0, ..., 0

It is also recommended to change the `nstagei` (section &NUMERICS) value to 15.

- **Eirene:**
 - NLSCL (section 1) should be set to "T" to ensure a good particle balance in Eirene
 - ILIIN (section 3a) for the "non-default surfaces" listed under the "lsns" above should be either 1 or 2 or -3 (not -1!) in order that the neutral fluxes be calculated by Eirene
 - The order of atomic species should be the same as in B2
 - The order of strata should be the same as in B2

Sample input files can be found in

`/afs/ipp/u/ank/pub/B2E/samples/particle_balance`
